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Review article



Non-suicidal self-injury in individuals with attention-deficit/hyperactivity disorder: A systematic review and Meta-analysis with Age- and Sex-stratified findings

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ABSTRACT

Background: Non-suicidal self-injury (NSSI) is a significant mental health concern that often co-occurs with attention-deficit/hyperactivity disorder (ADHD). Although growing evidence suggests a link between ADHD and NSSI, the prevalence and moderating factors of this association remain unclear.

Methods: We conducted a systematic review and meta-analysis to quantify the prevalence of NSSI in individuals with ADHD and to examine age- and sex-stratified differences across studies conducted in clinical, community, convenience, and population-based samples. A comprehensive search was performed across MEDLINE, EMBASE, and PsycINFO. Fourteen studies met inclusion criteria. Pooled prevalence estimates and odds ratios (ORs) were calculated using random-effects models. Subgroup analyses were performed by age group (adolescents vs. adults) and sex.

Results: The overall pooled prevalence of NSSI among individuals with ADHD was 27% (95% CI: 19%–37%), with similar rates among adolescents (28%) and adults (25%). Compared to individuals without ADHD, those with ADHD had twice the odds of engaging in NSSI (OR = 2.26, 95% CI: 1.69–3.00). Sex-stratified analysis, based on three eligible studies, suggested a higher risk in females with ADHD compared to males with ADHD (OR = 4.07, 95% CI: 3.09–5.36). High heterogeneity was observed across studies, potentially attributable to differences in diagnostic criteria, assessment tools, sample characteristics, and sociocultural contexts.

Conclusions: Individuals with ADHD show elevated rates of NSSI across developmental stages, with preliminary evidence indicating a potentially greater risk among females. However, these sex-specific findings should be interpreted cautiously due to the limited number of studies. Future research should investigate underlying mechanisms and longitudinal trajectories to better inform targeted interventions and prevention strategies. PROSPERO: CRD420251017948.

1. Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is a childhood-onset neurodevelopmental disorder that frequently persists into adolescence and adulthood. It is clinically characterized by core symptoms of inattention, hyperactivity, and impulsivity (Drechsler et al., 2020). Globally, the prevalence of ADHD among children and adolescents is estimated to range between 5 % and 7 %, with a higher diagnostic rate in males (Thomas et al., 2015). Beyond these core symptoms, individuals with ADHD often exhibit difficulties in emotional regulation, impaired executive functioning, and high rates of psychiatric comorbidities, such as mood disorders, conduct disorders, and substance use disorders (Gnanavel et al., 2019). These factors contribute to significant impairments in quality of life, academic performance, and social relationships (Danckaerts et al., 2010; French et al., 2024).

Non-suicidal self-injury (NSSI) is defined as any deliberate act of self-inflicted bodily harm without the explicit intention to die (Cipriano et al., 2017). This behavior is particularly prevalent during adolescence,

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with reported rates ranging from 13 % to 20 % across various populations (Farkas et al., 2024; Muehlenkamp et al., 2012). Common methods of NSSI include superficial cutting, scratching, burning, or ingestion of non-lethal substances (Klonsky et al., 2014). In the literature, this phenomenon is also referred to using a range of terminologies, such as self-injurious behavior, deliberate self-harm, and non-fatal self-injury. Adolescents who experience emotional dysregulation, impulsivity, or psychiatric comorbidities such as depression are at increased risk of engaging in NSSI (Andover and Morris, 2014; Niu et al., 2024). Given the overlap between the clinical features of ADHD and the risk factors associated with NSSI, a potential association between these conditions has been proposed (Allely, 2014).

Although several studies have reported an association between ADHD and self-injurious behaviors, particularly among adolescents with clinical or subthreshold symptoms, the evidence remains scattered and inconclusive (Austgulen et al., 2024; Balázs et al., 2018; Mulraney et al., 2021; Ojala et al., 2022). Many investigations are based on clinical or convenience samples, which may limit the generalizability of their findings. Moreover, there is a lack of comprehensive, quantitative synthesis regarding the frequency of NSSI in individuals with ADHD across developmental stages and by sex. This gap hinders the development of targeted prevention and intervention strategies. To address these limitations, the present study aims to conduct a systematic review and meta-analysis to estimate the pooled frequency of NSSI among individuals with ADHD, across studies conducted in clinical, community, convenience, and population-based samples; and to examine potential moderating factors, including age and sex, where data are available.

2. Methods

We conducted a systematic review with meta-analysis, following the guidelines set by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (Page et al., 2021).

2.1. Eligibility criteria

Studies were considered for inclusion if they assessed patients with an ADHD diagnosis and reported the frequency of NSSI. ADHD had to be identified through DSM- or ICD-based diagnostic criteria, structured or semi-structured clinical interviews, clinician-confirmed diagnoses, or validated rating scales applying established clinical cutoffs. NSSI was defined as any deliberate act of self-inflicted bodily harm without the explicit intention to die. Examples of NSSI included superficial cutting, scratching, burning, among others. We included cross-sectional studies, cohort studies, and secondary analyses of randomized controlled trials when they reported the proportion or frequency of NSSI among individuals with ADHD. Case reports, posters, and review articles were excluded, as were studies that did not clearly distinguish between NSSI and suicidal intentionality.

2.2. Search strategy

We conducted a systematic search of three electronic databases—MEDLINE, EMBASE, and PsycINFO—for articles published up to March 2025. Only studies published in English were considered for inclusion. The search strategy combined two main clusters of terms: one related to attention-deficit/hyperactivity disorder (e.g., “attention deficit hyperactivity disorder”, “ADHD”) and the other encompassing non-suicidal self-injury and its related constructs. To ensure comprehensive retrieval, the NSSI cluster included a range of synonymous and closely related terms such as “non-suicidal self-injury”, “self-injurious behavior”, “self-harm”, “deliberate self-harm”, and “non-fatal self-injury”. For example, the search strategy used in MEDLINE can be found in the supplementary material. Additionally, the reference lists of the included articles were manually reviewed to identify any additional relevant studies.

2.3. Study selection

Three reviewers independently screened the titles and abstracts of the retrieved articles. Full-text articles were reviewed when the eligibility criteria were unclear. Following this, each article selected in the screening phase was independently assessed in full by three reviewers for inclusion in the narrative analysis. In both phases, any disagreements were resolved through discussion. For inclusion in the quantitative analysis, studies were selected by consensus among the three reviewers.

2.4. Data extraction

Data from each selected article were independently extracted by three reviewers using a standardized spreadsheet developed by the authors. Any discrepancies in the extracted data were reviewed by the reviewers, and disagreements were resolved through discussion and consensus.

2.5. Statistical analysis

We conducted random-effects meta-analyses to estimate the pooled frequency of non-suicidal self-injury among individuals with ADHD and the corresponding 95 % confidence intervals (CI). A random-effects framework was selected a priori because substantial clinical and methodological heterogeneity was expected across studies, including differences in sampling frames (clinical, community, and population-based samples), diagnostic criteria for ADHD, assessment tools for NSSI, age composition, and sociocultural context. Under this model, true effect sizes are assumed to vary between studies, and both within-study and between-study variance are incorporated into the pooled estimates, allowing for more conservative and generalizable inferences compared to fixed-effect approaches (Barili et al., 2018).

For studies that included a comparison group without ADHD, we performed separate random-effects meta-analyses to estimate the odds of NSSI in individuals with ADHD relative to non-ADHD participants. To further explore potential sources of heterogeneity, we conducted predefined subgroup meta-analyses stratified by age group (adolescents versus adults) and by sex within the ADHD population, following established recommendations for subgroup analyses in meta-analytic research (Borenstein and Higgins, 2013; Choi and Kang, 2025).

Statistical heterogeneity was assessed using Cochran’s Q test, with a significance threshold of $P < 0.10$, and quantified using the I^2 statistic, which describes the proportion of total variability attributable to between-study heterogeneity rather than sampling error. Potential publication bias and small-study effects were evaluated only for meta-analytic models including more than ten studies, in accordance with methodological guidelines. These assessments involved visual inspection of funnel plot asymmetry and formal testing using Egger’s regression intercept (Page et al., 2021). All analyses were conducted using R statistical software.

3. Results

A total of 1514 articles were identified using our search strategy across the three databases. After removing duplicates, 1280 titles and abstracts were screened. At this stage, the primary reasons for exclusion were inappropriate outcomes, study populations, and study designs (e.g., case reports, case series, reviews, and guidelines). Following this process, 99 articles were selected for full-text review, of which 14 were included in the systematic review and meta-analysis. The study selection flowchart, along with reasons for exclusion, is presented in Fig. 1.

Most of the included studies had a cross-sectional design (8 studies), while 5 were longitudinal in nature and 1 study was a secondary analysis of a randomized clinical trial. Sample sizes varied widely, ranging from 74 to 10,957 participants, and included both adolescent and adult populations. The mean age ranged from 10 to 44.8 years, and the

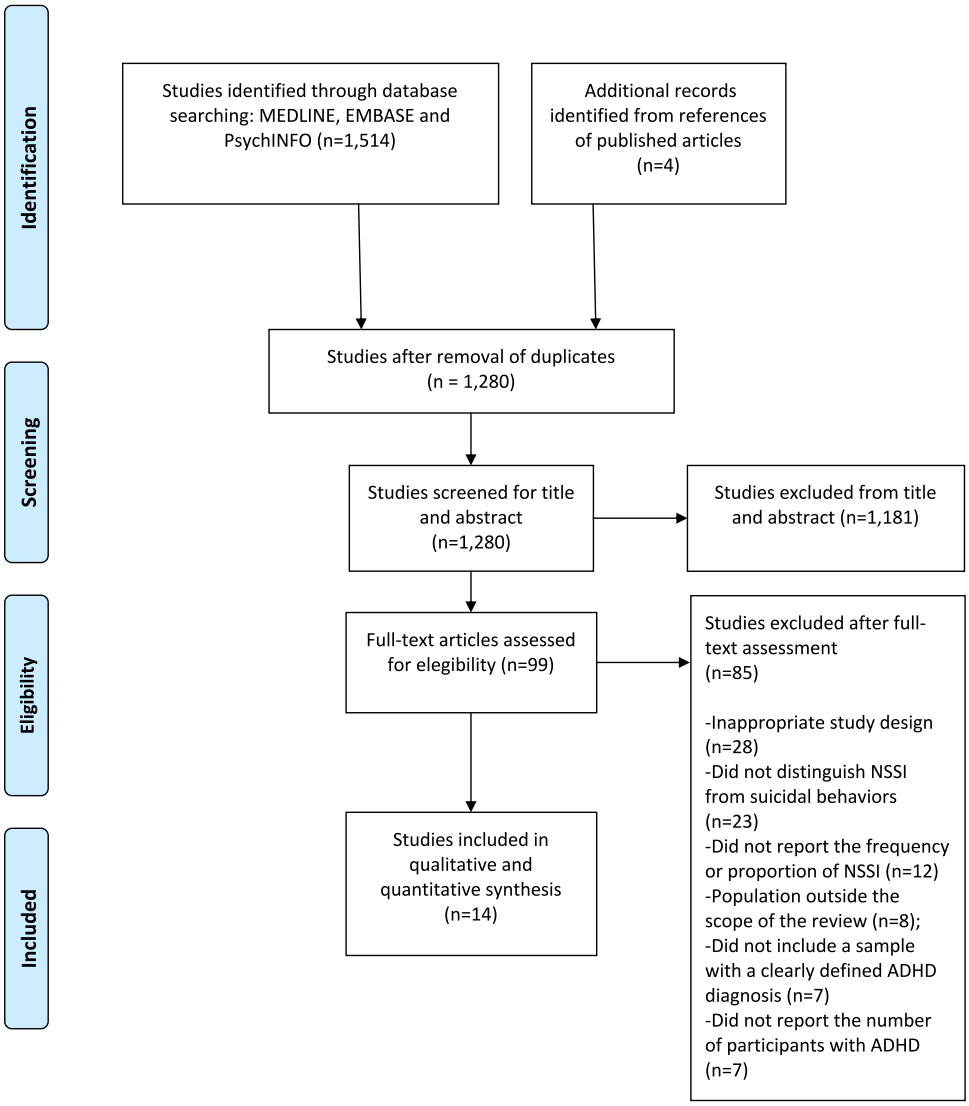


Fig. 1. PRISMA Flow Diagram of Study Selection Process for the Systematic Review and Meta-Analysis.

proportion of female participants varied substantially, from 11.8 % to 100 %. ADHD diagnosis was established through structured clinical interviews, validated self-report measures, or formal diagnostic criteria. NSSI was assessed using standardized instruments, clinical rating scales, or specific self-report questions regarding self-injurious behavior, with recall periods ranging from lifetime history to the past 12 months. The methodological characteristics of the included studies are summarized in Table 1.

3.1. *Frequency of NSSI in adolescents with ADHD*

Hurtig et al. (2012), using the Northern Finland Birth Cohort, found that adolescents with ADHD ($n = 104$) had significantly higher rates of NSSI (69 %) than controls (31 %, $n = 169$), based on K-SADS-PL) item "non-suicidal physical self-damaging acts," which captures behaviors such as cutting or burning without suicidal intent (Hurtig et al., 2012). Similarly, Kılıç et al. (2017), applying the SIBQ in Turkish adolescents, reported that 30 % of those with ADHD ($n = 30$) engaged in NSSI and 6.7 % attempted suicide, with higher trauma and dissociation scores than controls (Kılıç et al., 2017).

Balázs et al. (2018) examined 202 psychiatric inpatients in Hungary and showed that among those with ADHD ($n = 52$), 67.3 % engaged in NSSI. The association was statistically accounted for by co-occurring conditions, including affective disorder, suicidality, psychosis, and alcohol abuse in girls (Balázs et al., 2018). In line with these findings, Gamli and Tahiroglu (2018) studied 82 treatment-naïve ADHD adolescents and reported baseline NSSI frequency of 58.5 % (mean onset age 11.4). Adolescents with NSSI showed higher impulsivity scores, and those receiving methylphenidate exhibited scores in a more favorable direction compared with those not receiving medication, although impulsivity levels remained elevated (Gamli and Tahiroglu, 2018).

Population-based studies also confirmed increased risk, including the Northern Finland Birth Cohort reported by Hurtig et al. (2012), as well as studies by Austgulen et al. (2024); Kazda et al. (2022), and Mulraney et al. (2021). Mulraney et al. (2021), in 2653 Australian adolescents, found higher lifetime self-harm in subsyndromal ADHD (20.9 %) compared to ADHD (14.2 %) and controls (11.3 %) (Mulraney et al., 2021). Kazda et al. (2022), analyzing 393 ADHD adolescents and matched controls, showed reduced quality of life and a higher risk of self-harm (OR=2.53) (Kazda et al., 2022). Austgulen et al. (2024), with 9692 participants, observed self-harm frequency of 14.6 % in ADHD vs. 5.4 % in non-ADHD, with risk remaining significant after adjustment for age, sex, parental education, depressive symptoms, conduct problems, and familial history of self-harm or suicide attempts (OR=1.58) (Austgulen et al., 2024).

Finally, longitudinal evidence reinforces the association. Lin et al.

Table 1
Characteristics of the studies included in the systematic review.

Author (Year)	Location	Study Design	Sample Size Number	Age Group	Age, range and mean (yrs)	Female (%)	ADHD assessment	NSSI Assessment
Hinshaw et al. (2012)	USA	Longitudinal	198	Adult	Age range: 17–24; Mean 19.6.	100	DISC–IV	SIQ; NSSI in the past year.
Hurtig et al. (2012)	Finland	Longitudinal	273	Adolescent	Age range: 16–18.	57,9	SWAN	K-SADS-PL
Kimbrel et al. (2017)	USA	Cross Sectional	140	Adult	Age > 18; Mean 40.2	11.8	ASRS	DSHI; Lifetime history of NSSI.
Kiliç et al. (2017)	Turkey	Cross Sectional	207	Adolescent	Age range: 12–18; Mean 14.9.	36.7	KSADS-PL-T	A self-report questionnaire was developed by the authors to investigate frequency, duration and types of SIB over the life course.
Balázs et al. (2018)	Hungary	Cross Sectional	202	Adolescent	Age range: 13–18; Mean: 14.8.	71.4	M.I.N.I. Kid 2.0	DSHI
Gamli and Tahiroglu (2018)	Turkey	Longitudinal	82	Adolescent	Age range: 12–18.	47.6	K-SADS-PL	ISAS
Mulraney et al. (2021)	Australia	Cross Sectional	2653	Adolescent	Age range: 12–17.	48.4	DISC-IV	Specific question about NSSI lifetime and in the past 12 months.
Kazda et al. 2022	Australia	Longitudinal	786	Adolescent	Age range: 6–15; Mean:10.	27.8	SDQ	Specific question about NSSI in the past 12 months.
Olsson et al. (2022)	Swedish	Longitudinal	804	Adult	Age > 18; Mean: 33.	67.3	Based on ICD-10 criteria (F90.0–F90.9) or current use of ADHD medication.	Present to the emergency department due to self-harm or suicidal attempt.
Smari et al. (2023)	Iceland	Cross Sectional	10,957	Adult	Age range: 18–69. Mean: 44,8	100	Self-reported lifetime history of an ADHD diagnosis.	CIDI; Lifetime history of NSSI.
Di Salvo et al. (2024)	Turin	Cross Sectional	74	Adult	Age > 18; Mean: 30;	32.4	Clinical Interview (DSM-5-TR).	CSSRS (Specific item on the behavior subscale - Lifetime).
Austgulen et al. (2024)	Norway	Cross Sectional	9692	Adolescent	Mean: 17.4.	53	ASRS	CASE; Lifetime history of NSSI.
Lin et al. (2024)	Australia	Longitudinal	3696	Adolescent	14	48.6	Based on parental reports.	Self-report of any self-harm over the past 12 months.
Ergül and Ersöz Alan (2024)	Turkey	Cross Sectional	104	Adolescent	Age range: 12–18;	60.6	K-SADS-PL	Self-report of any self-harm.

ADHD: Attention-deficit/hyperactivity disorder; NSSI: Non-suicidal self-injury; DISC-IV: Parent-administered Diagnostic Interview Schedule for Children (4th ed); SIQ: Self-Injury Questionnaire; SWAN: Strengths and Weaknesses of ADHD and Normal Behavior Scale; K-SADS-PL: Schedule for Affective Disorders and Schizophrenia for School-Age Children-Present and Lifetime Version; ASRS: ADHD Self-Report Scale; DSHI: Deliberate Self-Harm Inventory; SIB: Self-injurious behavior; M.I.N.I. Kid 2.0: Hungarian version of the modified Mini International Neuropsychiatric Interview Kid; ISAS: Inventory of Statements About Self-Injury; SDQ: Strengths and Difficulties Questionnaire; CIDI: Composite International Diagnostic Interview; CSSRS: Columbia-Suicide Severity Rating Scale; CASE: Child and Adolescent Self-harm in Europe;.

(2024) followed 3696 youths and showed that ADHD at age 10 predicted a markedly higher risk of self-harm by age 14 (OR=25.14), partly mediated by depression and bullying (Lin et al., 2024). Ergül and Ersöz Alan (2024) further highlighted that Cognitive Disengagement Syndrome (CDS), particularly daydreaming and slowness, was independently associated with NSSI, alongside low self-esteem and emotion dysregulation (Ergül and Ersöz Alan, 2024).

3.2. Frequency of NSSI in adults with ADHD

Hinshaw et al. (2012) followed 140 girls with childhood ADHD and 88 controls over 10 years, finding significantly higher self-injury rates in early adulthood (50.6 % in ADHD-C vs. 19.0 % in controls). Risk was particularly elevated in the combined subtype and persisted after adjustment for comorbidities, IQ, and socioeconomic factors (Hinshaw et al., 2012). Extending to male populations, Kimbrel et al. (2017) studied 140 Iraq/Afghanistan-era veterans and reported that ADHD remained strongly associated with NSSI (AOR = 3.51, $p = 0.031$) even after adjusting for PTSD, depression, and demographic variables, with alcohol use disorder as the only other significant factor (Kimbrel et al., 2017).

Further evidence in adults highlights the role of impulsivity and comorbidities. Olsson et al. (2022), in a six-month follow-up of 804 adults, found that those with ADHD were more likely to engage in impulsive self-harm, particularly linked to emotional dysregulation and relationship difficulties, though lethality did not differ from controls (Olsson et al., 2022). Smari et al. (2023), focusing on women with cardiometabolic conditions, showed that co-occurring ADHD was associated with markedly higher rates of depression (PR = 2.38), bipolar disorder (PR = 4.81), and PTSD (PR = 2.78), as well as increased risk of self-harm (PR = 1.47) and suicide attempts (PR = 2.37) (Smari et al., 2023).

Finally, Di Salvo et al. (2024) assessed suicidality in ADHD adults using the Columbia-Suicide Severity Rating Scale (C-SSRS). Lifetime frequency was high in this adult sample (mean age of approximately 30 years), with 59.5 % reporting suicidal ideation, 16.2 % severe ideation, 9.5 % suicidal behavior, and 10.8 % engaging in NSSI, underscoring the chronic vulnerability to self-injury across the lifespan [28].

3.3. Meta-analysis

Fourteen studies were included in the meta-analysis evaluating the frequency of NSSI in patients with ADHD. The pooled frequency of NSSI was estimated at 27 % (95 % CI: 19 %–37 %, $I^2 = 95$ %) (Fig. 2). The funnel plot suggested visual asymmetry, and Egger's regression was statistically significant ($p = 0.02$), indicating potential publication bias or small-study effects (Fig. 3).

Subgroup analyses were conducted based on the age group in which

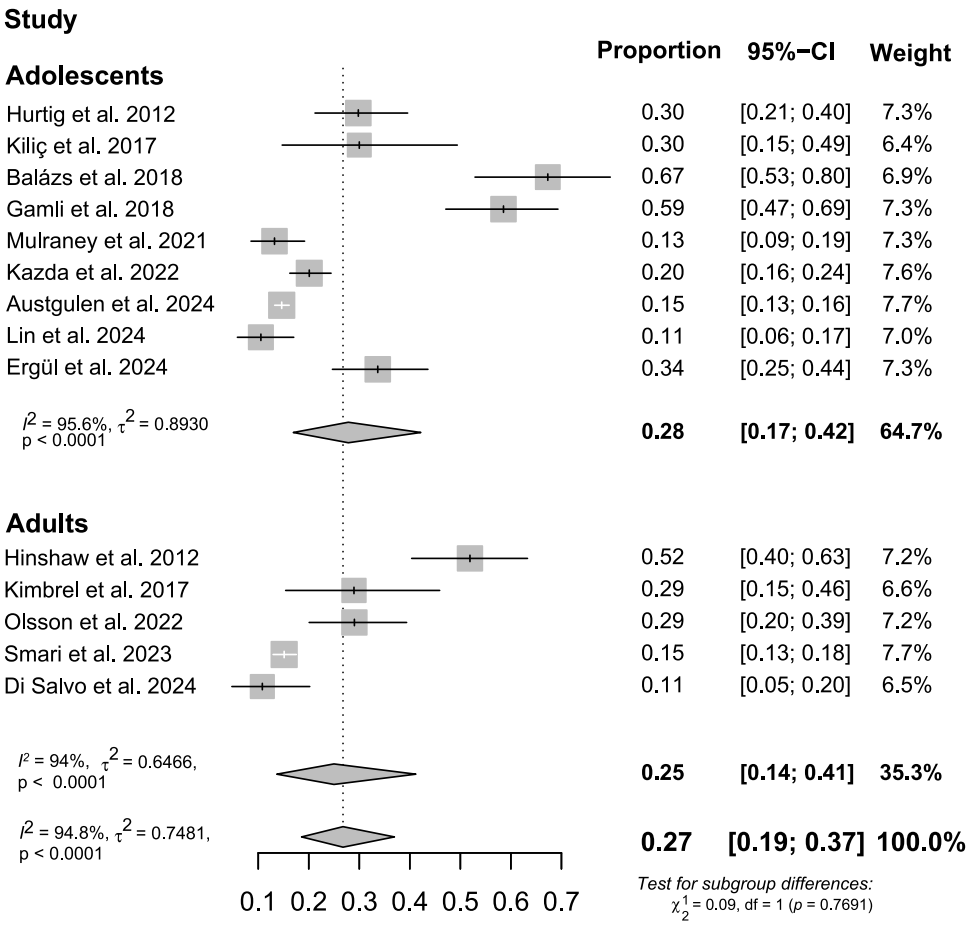


Fig. 2. Forest plot of the pooled frequency of non-suicidal self-injury (NSSI) among individuals with attention-deficit/hyperactivity disorder (ADHD) across the included studies, with subgroup analyses stratified by age group (adolescents vs. adults).

symptoms were assessed. Among the nine studies that evaluated NSSI in adolescent populations, the frequency was 28 % (95 % CI: 17 %–42 %, I^2 = 96 %). For the five studies that assessed adult populations, the frequency was 25 % (95 % CI: 14 %–41 %, I^2 = 94 %). No studies were found that reported frequency in children.

Nine studies, comprising approximately 28,600 participants, allowed for comparisons between individuals with ADHD and control groups. The meta-analysis revealed that patients with ADHD exhibited a higher frequency of NSSI compared to controls (pooled OR = 2.26, 95 % CI: 1.69–3.00, I^2 = 80 %). In subgroup analyses, five studies allowed for this comparison in adolescent populations, yielding a pooled OR of 2.40 (95 % CI: 1.64–3.53, I^2 = 74 %), while four studies focused on adult populations, with a pooled OR of 2.08 (95 % CI: 1.26–3.43, I^2 = 69 %) (Fig. 4).

3.4. Sex

Across the reviewed studies, sex was examined as a potential factor in the association between ADHD and self-injurious outcomes. Kılıç et al. (2017) identified female sex as a significant predictor of suicide attempts, but not of non-suicidal self-injury (NSSI) (Kılıç et al., 2017). Balázs et al. (2018) reported higher rates of NSSI among girls with ADHD, with 71.4 % of those presenting both conditions being female (Balázs et al., 2018). Austgulen et al. also found that female adolescents in the ADHD group were more likely to engage in self-harm, particularly those with severe inattention and hyperactivity/impulsivity symptoms, as well as higher levels of depressive symptoms (Austgulen et al., 2024). In contrast, Lin et al. (2024) observed higher rates of suicidal thoughts/attempts and self-harm among boys with ADHD, with no increased risk identified among girls with the condition (Lin et al., 2024).

A meta-analysis was conducted using data from three studies that reported NSSI frequency by sex in adolescents with ADHD. The pooled OR was 4.07 [95 % CI: 3.09–5.36], indicating higher odds of NSSI among girls compared to boys within the ADHD population. Heterogeneity was low (I^2 = 0 %, τ^2 = 0.00, p = 0.5), indicating consistency in effect estimates across the included studies (Fig. 5).

4. Discussion

In this study, we conducted a systematic review and meta-analysis to explore the frequency of NSSI among individuals with ADHD. Across fourteen studies including adolescents and adults, approximately 27 % of individuals with ADHD reported a history of NSSI, notably higher than the 18–22 % typically found in the general population (Lim et al., 2019; Swannell et al., 2014). Moreover, in a comparative analysis comprising over approximately 28,600 participants, individuals with ADHD demonstrated roughly twice the odds of engaging in NSSI compared to those without the disorder (pooled OR = 2.10, 95 % CI: 1.65–2.67). These findings suggest a meaningful association between ADHD and self-injurious behaviors and highlight the importance of systematically evaluating NSSI risk in clinical settings when working with this population.

The heterogeneity observed across the included studies was substantial, particularly in the pooled frequency estimates (I^2 > 90 %). This variability likely reflects a combination of methodological and contextual differences among the studies. Variations in the diagnostic procedures for ADHD were evident, with some studies relying on structured clinical interviews (such as the K-SADS-PL or DISC-IV), others using self-

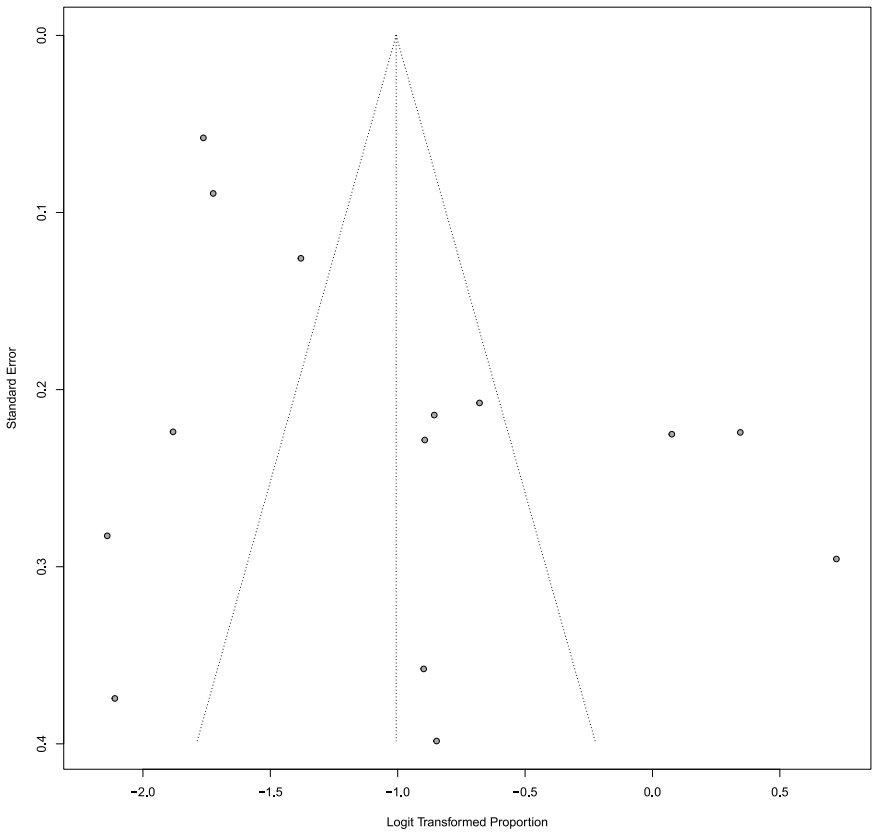


Fig. 3. Funnel plot assessing potential publication bias and small-study effects in the meta-analysis of non-suicidal self-injury (NSSI) frequency among individuals with attention-deficit/hyperactivity disorder (ADHD).

report instruments like the ASRS. Similarly, assessments of NSSI differed considerably, encompassing both validated instruments (e.g., DSHI, ISAS) and less standardized measures such as single-item self-reports or clinical interviews with differing recall periods. Cultural and contextual factors may also have played a role, given that the studies were conducted in a wide range of sociogeographic settings, potentially influencing how self-injurious behaviors are reported, perceived, and interpreted. Additionally, differences in sample characteristics—particularly between clinical populations and community-based cohorts—could have contributed to the variability, as clinical samples often include individuals with greater symptom severity, higher rates of comorbidity, and more pronounced functional impairment. Furthermore, although funnel plot inspection and Egger's regression suggested asymmetry, these findings should be interpreted with caution due to the limited number of studies and the high degree of heterogeneity, which may produce apparent asymmetry independent of publication bias or small-study effects.

The association between ADHD and NSSI likely reflects a complex clinical scenario shaped by multiple interacting psychopathological mechanisms. Rather than a direct or singular pathway, several underlying factors may contribute to the increased vulnerability to self-injury observed in individuals with ADHD. Impulsivity and emotional dysregulation, both core features commonly associated with ADHD, have been consistently implicated as proximal risk factors for NSSI (Cassels et al., 2022; Wolff et al., 2019). These traits may impair an individual's ability to tolerate distress, leading to maladaptive coping strategies such as self-injury. Additionally, interpersonal difficulties, including peer rejection, social isolation, and family conflict, are frequently reported in youth with ADHD (Sodano et al., 2021) and may further heighten emotional distress and contribute to NSSI as a form of affect regulation or social signaling (Muehlenkamp et al., 2013). Academic underachievement and school failure, often experienced by individuals with ADHD, can erode self-efficacy and fuel a sense of hopelessness (Polderman et al., 2010). Finally, low self-esteem and negative self-concept—which may develop over time through repeated experiences of failure and criticism—have been identified as important psychological vulnerabilities associated with both ADHD and NSSI (Forrester et al., 2017; Mazzone et al., 2013). These converging pathways underscore the need for a multidimensional and developmentally informed understanding of risk in this population.

An additional and particularly novel finding emerging from this review is the association between Cognitive Disengagement Syndrome (CDS) and NSSI among adolescents with ADHD (Ergül and Ersöz Alan, 2024). CDS, characterized by symptoms such as mental fogginess, daydreaming, and slowed cognition, represents a non-traditional attentional profile that is partially distinct from the classical inattention symptoms captured in standard ADHD criteria (Becker et al., 2023). The observed link between CDS and self-injurious behavior suggests that attentional disengagement and cognitive hypoactivity may constitute an underrecognized pathway to emotional dysregulation and maladaptive coping strategies, including NSSI. However, recent evidence from a community-based sample of early adolescents found that CDS was not independently associated with NSSI after adjusting for ADHD symptom severity (Thornton et al., 2025), suggesting that the contribution of CDS to self-injury may not extend beyond its overlap with core ADHD symptoms. Taken together, these divergent findings highlight that the role of CDS in self-injury remains uncertain and may vary according to population characteristics, measurement methods, or co-occurring psychopathology, reinforcing the need for further research to clarify whether CDS represents an independent risk factor or reflects broader attentional and emotional vulnerabilities within ADHD.

Psychiatric comorbidities appear to play a crucial moderating role in the relationship between ADHD and NSSI. Several of the studies included in this review identified associations with co-occurring

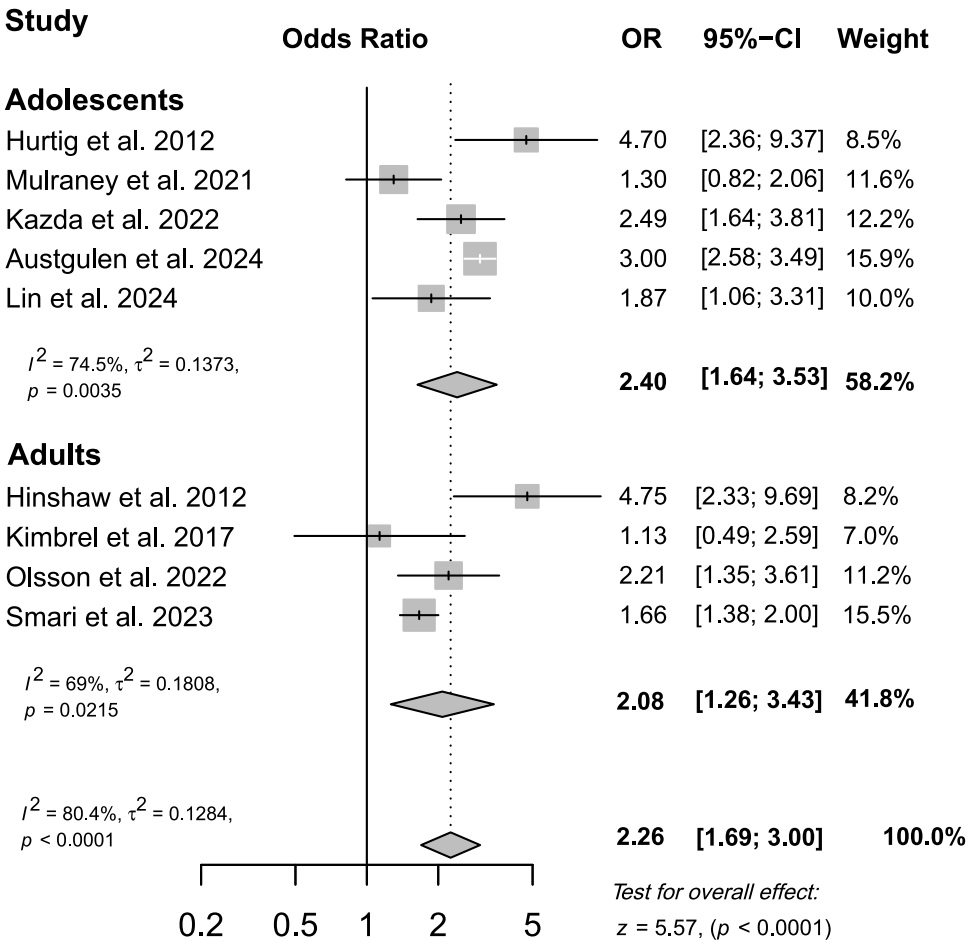


Fig. 4. Forest plot comparing individuals with attention-deficit/hyperactivity disorder (ADHD) and non-ADHD control groups, showing the association between ADHD and non-suicidal self-injury (NSSI) using odds ratios as the measure of effect.

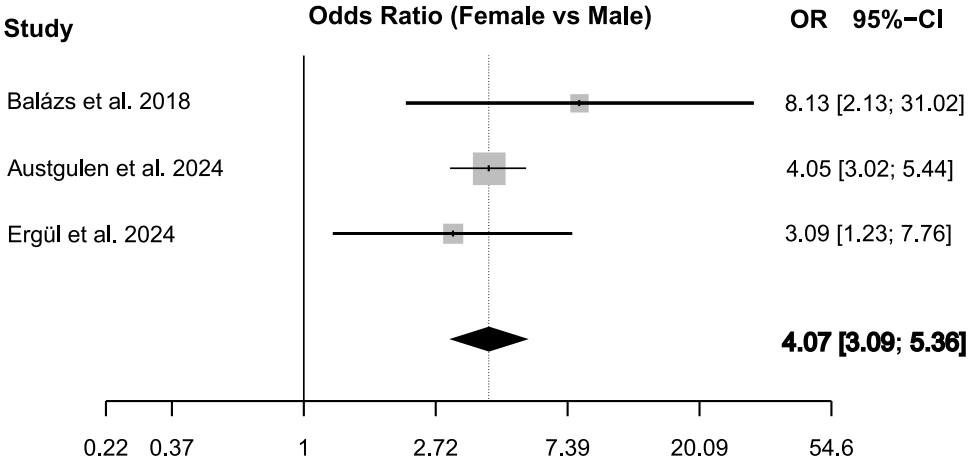


Fig. 5. Forest plot comparing females and males within the attention-deficit/hyperactivity disorder (ADHD) population, presenting odds ratios for non-suicidal self-injury (NSSI) to evaluate sex-related differences in risk among individuals with ADHD.

conditions—particularly depressive disorders, anxiety disorders, and PTSD—in relation to the likelihood and severity of self-injurious behaviors (Guan et al., 2024). These comorbidities may function as potential moderators, contributing to variations in risk across subgroups of individuals with ADHD, and they may also represent possible mediating pathways involving emotional dysregulation, hopelessness, rumination, and social withdrawal, all of which are closely linked to NSSI (Kraemer et al., 2001; Wu et al., 2023). Moreover, individuals with ADHD who experience such comorbidities may be at heightened risk due to cumulative functional impairments and reduced access to adaptive coping strategies (Gjervan et al., 2012).

In the age-stratified analysis, we observed a modest difference in NSSI frequency between adolescents (28 %) and adults (25 %) with ADHD. Although this difference was not statistically significant, it suggests that the risk of self-injurious behavior persists across developmental stages, rather than being confined to adolescence. This finding is particularly noteworthy given that prior studies have reported considerably lower rates of NSSI in the general adult population—typically ranging from 2.7 % to 5.9 % (Christoffersen et al., 2015; Klonsky et al., 2014). One possible explanation lies in the chronic nature of impulsivity, a core feature of ADHD that often endures into adulthood despite improvements in other symptom domains (Lopez et al., 2015). Persistent impulsivity, particularly when coupled with unresolved emotional dysregulation or ongoing functional impairments, may continue to predispose individuals to NSSI as a maladaptive coping strategy (Allen et al., 2019). Additionally, adults with ADHD may have accumulated long-standing difficulties in interpersonal relationships, occupational functioning, and self-concept, which may further sustain vulnerability to self-harm (Barkley and Murphy, 2010; Safren et al., 2010). These findings highlight the need to recognize and address NSSI risk not only in youth but also in adult clinical settings, where it may be underdetected or misattributed to other psychiatric conditions.

Sex emerged as a potentially important factor in the association between ADHD and NSSI. In our meta-analysis, females with ADHD were estimated to be four times more likely to engage in NSSI compared to males with the same diagnosis (OR = 4.07, 95 % CI: 3.09–5.36); however, this sex-stratified estimate was based on only three studies and should therefore be interpreted as preliminary. The observed disparity appears greater than that typically reported in the general population and may reflect the interaction between ADHD-related vulnerabilities and sex-specific risk factors (Moloney et al., 2024). Females with ADHD may be more prone to internalizing symptoms such as depression and anxiety, which are well-established correlates of NSSI and are frequently comorbid with ADHD in females (Siddiqui et al., 2024). A clinical case-control study found that women with adult ADHD exhibited a markedly increased likelihood of suicidal ideation compared to female controls (OR = 25.0, 95 % CI: 2.98–200.0), with “problems with self--concept” identified as the most strongly associated psychopathological domain (Kakuszi et al., 2018), suggesting that negative self-perception and internalized self-criticism may be relevant pathways. Taken together, these findings underscore the need for sex-sensitive assessment and intervention strategies in clinical settings, while highlighting the importance of further research specifically designed to investigate sex differences in NSSI among individuals with ADHD.

Several limitations of this study should be acknowledged. First, the high degree of heterogeneity observed across studies may limit the generalizability of the pooled estimates and reflects variability in methodological designs, sample characteristics, and measurement tools. Second, there was considerable variation in the instruments used to diagnose both ADHD and NSSI, ranging from structured clinical interviews to self-report questionnaires, which may introduce inconsistencies in case identification and outcome reporting. Third, the possibility of publication bias cannot be ruled out, as studies reporting significant associations are more likely to be published and indexed in databases, potentially inflating the observed effect sizes. Fourth, we identified a notable gap in the literature regarding NSSI frequency in children with ADHD, as no studies included in our meta-analysis specifically addressed this age group. This absence represents an important area for future research, particularly given that early identification and intervention may be crucial in preventing the onset and persistence of self-injurious behaviors.

In conclusion, this study addresses an important gap in the literature by delivering a quantitative synthesis of the frequency of non-suicidal self-injury among individuals with ADHD—a population known to experience elevated emotional dysregulation, impulsivity, and psychiatric comorbidity. Our findings indicate that approximately one in four individuals with ADHD have engaged in NSSI, and that the risk is nearly twice as high compared to those without the disorder, supporting the notion of overlapping vulnerabilities between ADHD and self-injurious behavior. Stratified analyses suggest that this elevated risk spans across age groups, with preliminary evidence pointing to a potentially greater vulnerability among females; however, these sex-specific findings should be interpreted cautiously as they are based on a limited number of studies. Taken together, these results underscore the importance of incorporating routine screening, early detection, and tailored interventions for NSSI within the broader clinical management of ADHD, while highlighting the need for further research focused on sex differences and developmental pathways.

Potential conflicts of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Liliane Smaniotto: Writing – original draft, Methodology, Conceptualization. **Arthur Tolentino:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Felipe M.R. Barros:** Data curation. **Eduardo Barreto:** Data curation. **Barbara Colnaghi:** Data curation. **Lais Ker:** Data curation. **Eloisa Helena Rubello Valler Celeri:** Writing – review & editing. **Osmar Della-Torre:** Writing – review & editing. **Renata C.S. Azevedo:** Writing – review & editing, Conceptualization.

Declaration of competing interest

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.psychres.2026.116975.

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